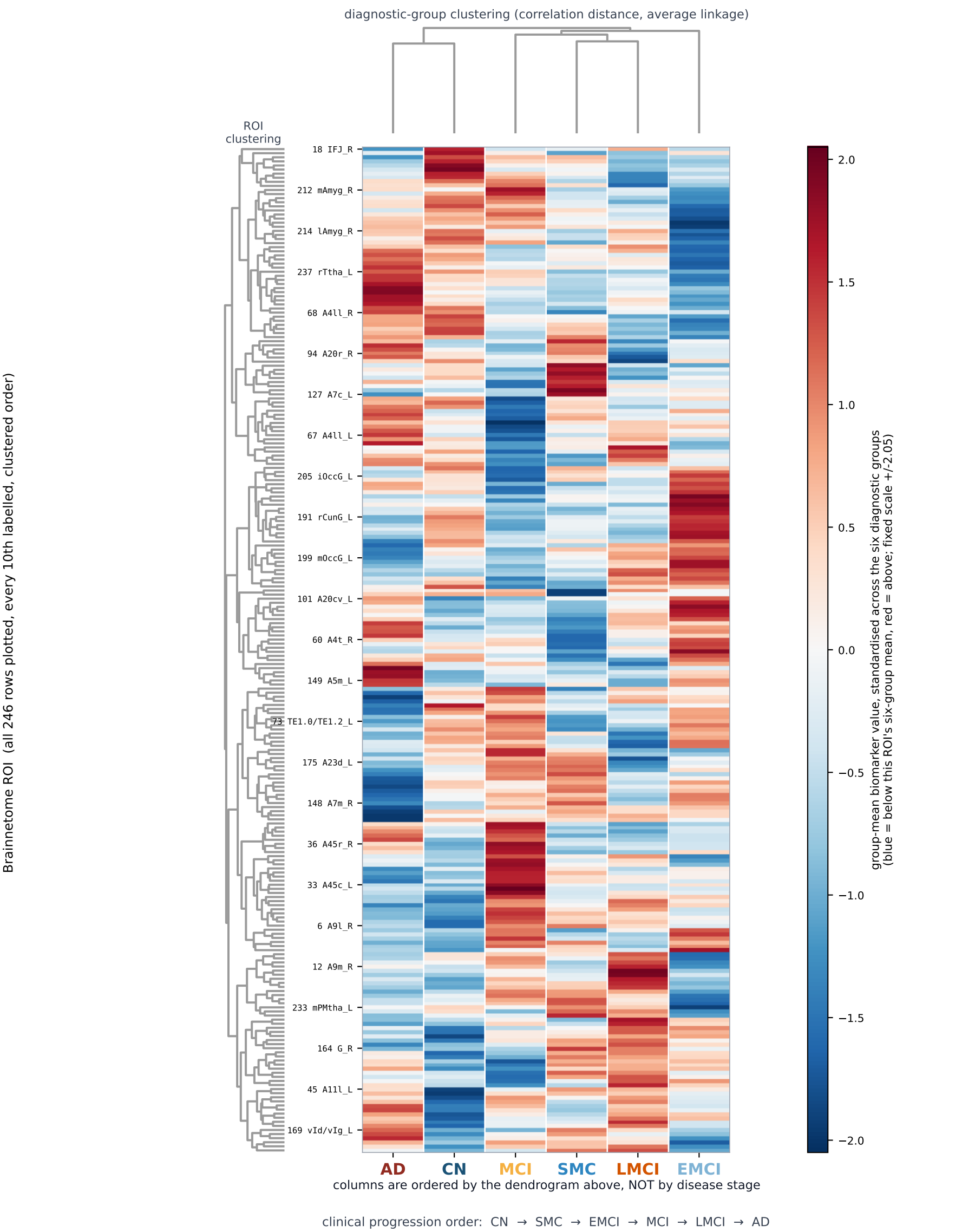
# Hierarchical Clustering of Alzheimer's Diagnostic Groups
# Regional homogeneity (mReHo)

cells are the MEAN mReHo of the subjects in each diagnostic group (n = 20/20/20/16/20/20), standardised across the six groups per ROI (ddof = 1)
groups clustered by correlation distance + average linkage over the 246-D profile; ROIs by euclidean distance + average linkage

group P-score = -log10(q) from the MULTIVARIATE stage-vs-rest permutation test (24-test family)

*max ROI P = largest ROI-level P-score for that group -- DESCRIPTIVE, not a test | both are far below the 1.301 significance threshold*

| | AD | CN | MCI | SMC | LMCI | EMCI |
|---|---|---|---|---|---|---|
| **group P-score** | **0.738** | **0.339** | **0.559** | **0.119** | **0.119** | **0.856** |
| *max ROI P* | 0.928 | 0.524 | 0.498 | 0.524 | 0.353 | 0.928 |




clinical progression order: CN → SMC → EMCI → MCI → LMCI → AD

bootstrap stability of the branches (1000 resamples of subjects within groups): {CN,AD} 36%; {SMC,LMCI} 16%; {SMC,EMCI,MCI,LMCI} 12%

Heatmap shows BIOMARKER ACTIVITY; the top row shows STATISTICAL SIGNIFICANCE. They are different quantities. The tree clusters on biomarker activity.

*Descriptive only. Branch heights are distances, not p-values. No group reaches q < 0.05 and no ROI x stage test in this project does either.*